

Kindled — Adversarial QA Agent

Multi-Agent AI for Game Development · Assignment #09 · Aaron Low · 24 Aug 2026
Game: Kindled (Godot 4.7, godot/) · Agent: godot/scripts/Adversary.gd · Runner: Scripts/godot-run.ps1 - Adversary · Reports: docs/qa/

Rubric map

Criterion	Pts	Where the evidence is
Findings	/ 4.0	Six real bugs, each named to a file:line and mechanic (Army.slots stagger, Battle._hitstop re-arm, slash tween / hit() await on freed units, juggle re-pop, enemy past the lens). Section "What it found".
Agent Logic	/ 3.0	15 hostile behaviors cycled on a seeded RNG through the real input map plus direct calls; "broken" = 14 explicit invariants the game's own code claims. Section "How the agent tries to break things".
Structured Report	/ 2.0	adversary_<seed>.json / .csv per run with location {scene, wave, system, unit_type, wx, wd}, error_type, game_context {behavior, magic, army, fps...}, count, screenshot; engine stderr folded in. Excerpt in the appendix.
ReadMe	/ 1.0	docs/qa/README.md — what it found, and "Was I surprised?". Reproduced in full below.

Committed runs

	Run	Findings	Engine errors	Error types
before fixes	adversary_11	5	89	boundary_break, exploit
before fixes	adversary_23	5	28	boundary_break, exploit
before fixes	adversary_7	4	145	boundary_break
after fixes	adversary_11	0	0	—
after fixes	adversary_23	0	0	—
after fixes	adversary_7	0	0	—

Adversarial QA agent — Kindled (Godot build)

Assignment #09. The agent lives inside the game as an autoload, `godot/scripts/Adversary.gd`, and is inert unless launched with `--adversary=<scene>,<seconds>,<seed>`.

```
pwsh Scripts\godot-run.ps1 -Adversary battle,60,7
```

That runs the real game binary against the real project folder (same code the web export ships), lets the agent loose for 60 s, then copies `adversary_<seed>.json / .csv / .png` here and folds the engine's own `SCRIPT ERROR` lines into the JSON as `engine_errors`.

How the agent tries to break things

Every 0.6 s it picks one of 15 behaviors (seeded RNG, so a run is repeatable) and holds it:

behavior	what it does
move_random, wall_hug, corner_hold	holds WASD through the real input map, including 2–4 s pinned into a wall or corner
cursor_warp, siphon_hold	mouse off-screen / hold RMB siphon with the cursor anywhere
spell_spam, swap_spam	cast Z/X/C or cycle the army (Q/E) every frame
teleport_hero	writes the hero far outside the hall, then watches the clamp
magic_flood, magic_drain	1e6 magic (relic thresholds) / 0 magic then cast everything (cost gate)
horde	drops 60 enemies right on the front line
launch_all	knocks every unit on the field into the air
kill_front, kill_hero	instakill the whole front line / set hero HP to 0 (lose path)
jump_phase	dev-jumps to a random phase mid-fight (scene swap while things are in flight)

"Broken" is defined as a rule the game's own code claims and then fails to keep. Checked every frame: hero inside its clamp, `Game.magic >= 0`, unit hp in `0..max_hp`, dead flag matches hp, `front[]` lanes point at live units that agree on their lane, `pool >= 0`, every unit between the hall walls (`|wx| <= HALL_HALF`) and inside `0..SPAWN_D+200` depth, `air_h >= 0`, no NaN, transient states end inside a budget (RETREAT 12 s, ADVANCE 15 s, charge 10 s, airborne 15 s), `< 1500` units, `fps >= 20`, and `Engine.time_scale` back to 1 within a second. Repeats of the same (type, system, unit type) are folded into one finding with a count.

Report fields: `error_type`, `location {scene, wave, system, unit_type, team, state, wx, wd}`, `game_context {behavior, magic, hero_hp, army_type, units, enemies, fps, held_inputs}`, `detail, t, count, screenshot` — the frame the rule first broke on, saved as `adversary_<seed>_f<id>.png` next to the report. `adversary_<seed>.png` is the final frame. Engine errors carry no screenshot (they come from Godot's stderr).

The seed fixes the behavior order; frame timing does not, so a 60 s run reproduces bugs 1–3 every time and bugs 4–6 in some runs. Findings below cite the run they appear in.

What it found (seeds 7, 11, 23 — 60 s each)

1. **Rank block sits through the hall wall** — `Army.slots` (godot/scripts/Army.gd:36) staggers odd rows by half a file: $-180 + 5 \times 72 + 36 = 216$, past `HALL_HALF = 200`. Every wave, every ally type in the outer file (veteran, hammer, halberdier, vet_ranged) stands 9–22 units inside the drawn wall. All three runs, from the first checked frame. Fix: subtract the stagger from `half` (or shrink `RANK_HALF` by $dx/2$). Shot: `adversary_7_f1.png` — outer helmets drawn over the bricks on both sides; `adversary_23_f4.png` same after a swap.
2. **Hit-stop exploit** — `Battle._hitstop` (godot/scripts/Battle.gd:382) sets `Engine.time_scale = 0.05` and every army swap re-arms it. Mashing Q/E holds the whole world at 5 % speed for as long as the player keeps mashing; the agent measured it held for the full spam tick every time. Fix: rate-limit swaps, or don't re-arm hit-stop while one is live. Shot: `adversary_11_f5.png` / `adversary_23_f5.png` (the swap toast is up; the slowdown itself is in the `detail` field, not visible in a still).
3. **Slash tween outlives its unit** — `SCRIPT ERROR: Invalid access to property or key 'team' on a base object of type 'Nil' at Battle.gd:332, 7–35× per run` (plus up to 109 `Lambda` capture ... was freed). The slash effect's tween callbacks capture attacker `a` and target `t`; when either dies/gets freed mid-clip (`kill_front`, wave reset, swap) the `lambda` runs on null. Fix: `if not is_instance_valid(a): return` at the top of the callback.
4. **Ranged hit after its attacker died** — `SCRIPT ERROR: Invalid access to property or key 'type' on a base object of type 'Nil' at Battle.gd:295 in hit(): the arrow branch awaits 0.12 s and then checks is_instance_valid(t) but not a`. Seen in two runs (3× and 12×). Same one-line fix.
5. **Perpetual juggling** — wave 2–3: units of both teams airborne for 15 s+. `Unit.take` adds `POP` on every hit while airborne and `_slash/_vortex` re-launch airborne foes, so a crowd under a whirl field never lands. Allies caught in it can't retreat or advance (`air_h > 0` branch skips movement). Shot: `juggle_seed7_earlier_run.png` — the crowd tumbling above the ceiling line, hero HP untouched.
6. **Enemy walks behind the lens** — a `mace_undead` at depth `-0` after a horde. Enemies that push past the hero keep walking toward `wd = 0` and beyond instead of being recalled or hitting the hero. Seen once; not in the committed runs.

Not found (the invariants held): hero clamp (teleporting 3000 units out snaps back next frame), magic never went negative through the cost gate, no `pool` underflow, no duplicate/dead `front[]` refs, no NaN, no fps collapse even with 4× hordes on top of each other.

After the fixes (same three seeds, re-run)

All six were fixed in the game and the agent re-run on seeds 7, 11, 23: **0 findings, 0 engine errors** in each 60 s run (reports in `after/`), and the four existing self-check probes (`swap`, `hammer`, `whirl`, `charge`) still pass. The before-fix reports stay in this folder as the evidence for the findings above.

bug	fix
1 rank block through the wall	<code>Army.slots</code> spaces <code>per_row - 0.5</code> files across <code>half</code> , so the staggered outer file lands on the edge, not past it; <code>Unit</code> also clamps <code>wx</code> inside <code>HALL_HALF</code> every frame
2 hit-stop exploit	<code>_set_army</code> accepts a swap every 0.5 s; <code>_hitstop</code> never extends a live stop; the restore timer calls <code>Engine.set_time_scale</code> with no <code>lambda</code> and <code>Battle._exit_tree</code> resets it, so a stop can't outlive the scene
3 slash tween on freed unit	the clip captures <code>a.team</code> and <code>t's</code> instance id by value

bug	fix
4 hit() after await	is_instance_valid(a) alongside the t check; the echo-relic callback resolves its unit through instance_from_id
5 juggling	kept — it's the design. Unit._air_cap() caps air_h at the height where the sprite top would leave the frame at that depth, so a juggled crowd bounces off the top edge instead of vanishing. Sky-drop entrances are exempt
6 enemy behind the lens	ENEMY_MIN_D = 130: the treadmill floors enemies at the hero's feet; whirl dodge-darts clamp to their slot depth so allies stop drifting backwards

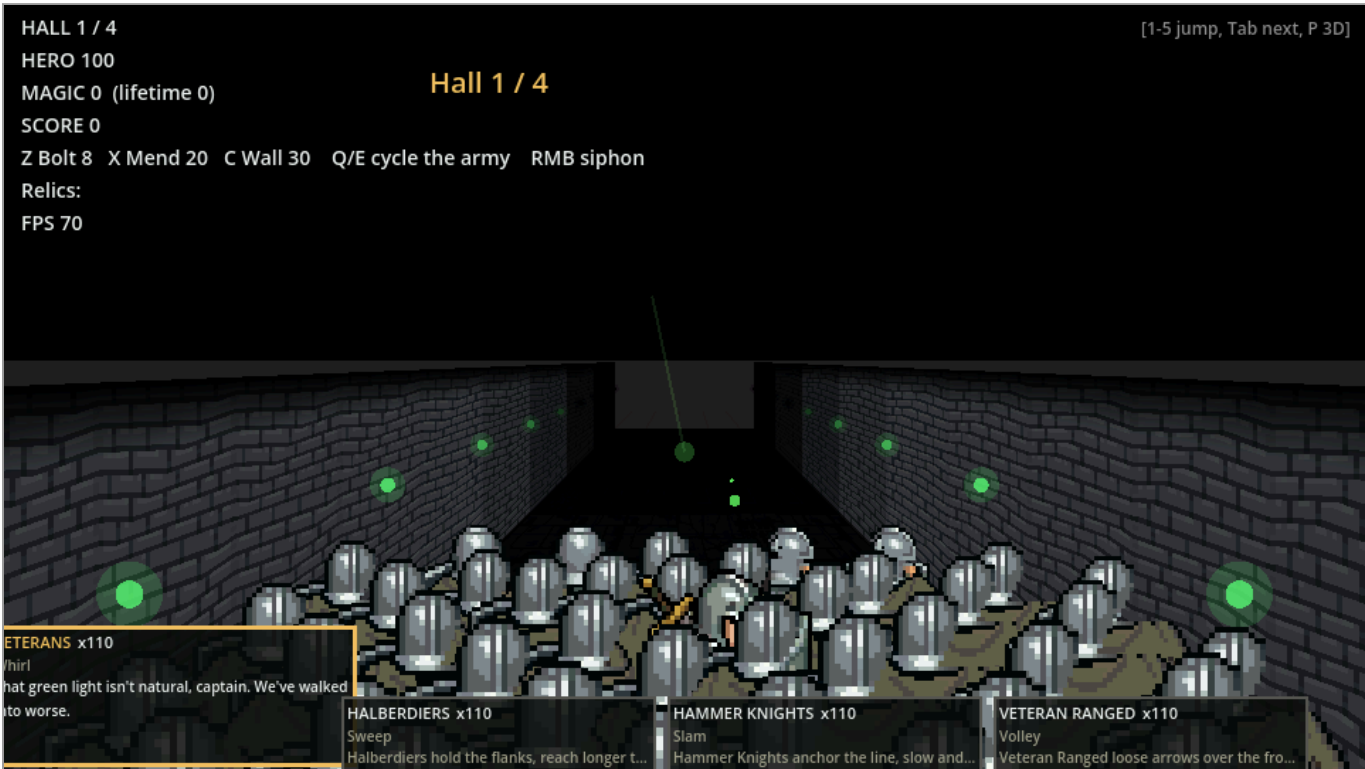
The agent's rules changed with the design: the "airborne 15 s" budget became "an airborne sprite's top stays on screen", and the depth floor became ENEMY_MIN_D for enemies. Two things the re-run also caught in the agent itself: _process was on by default, so the agent was running in ordinary launches too (fixed with set_process(false) in _ready), and the runner treated a parse error as "0 findings" (now a red failure).

Was I surprised?

Yes, twice. The wall bug is in the very first second of a normal game and had been in every screenshot for weeks — the walls are drawn at HALL_HALF and the outer file just stands in them; nobody looked because the crowd reads fine. And the hit-stop exploit was found by accident: the agent's own clock crawled because the game slowed *itself* down under swap spam, which is exactly the kind of thing a player mashing keys will do on the first try. The slash-tween null access was the expected one — captured object lifetimes in tween callbacks are a known Godot foot-gun — but 7–35 hits a minute says it fires in ordinary play too, not just under abuse. The floating crowd was not on the list at all; the 15 s budget was a guess and it tripped.

Screenshots

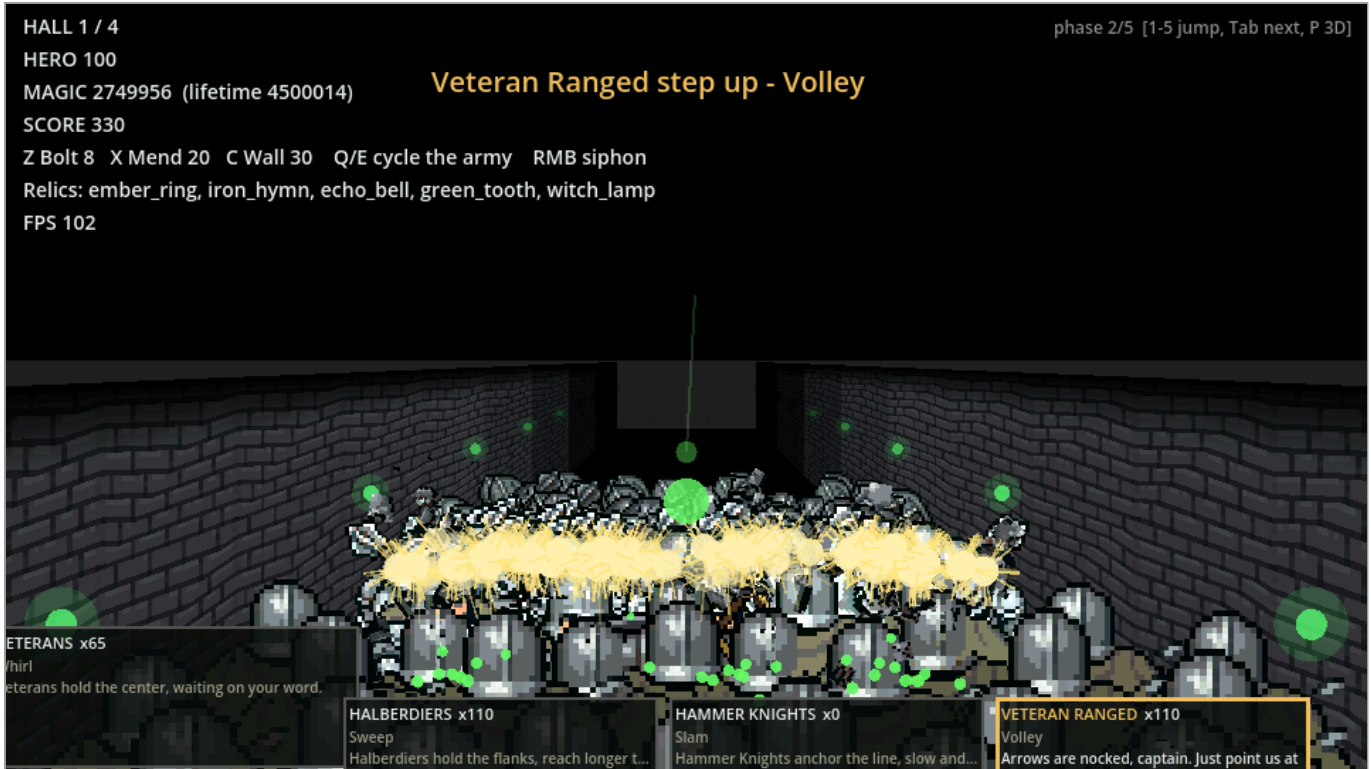
Each finding stores the frame it first broke on (screenshot field). Pixel art shown at native scale.



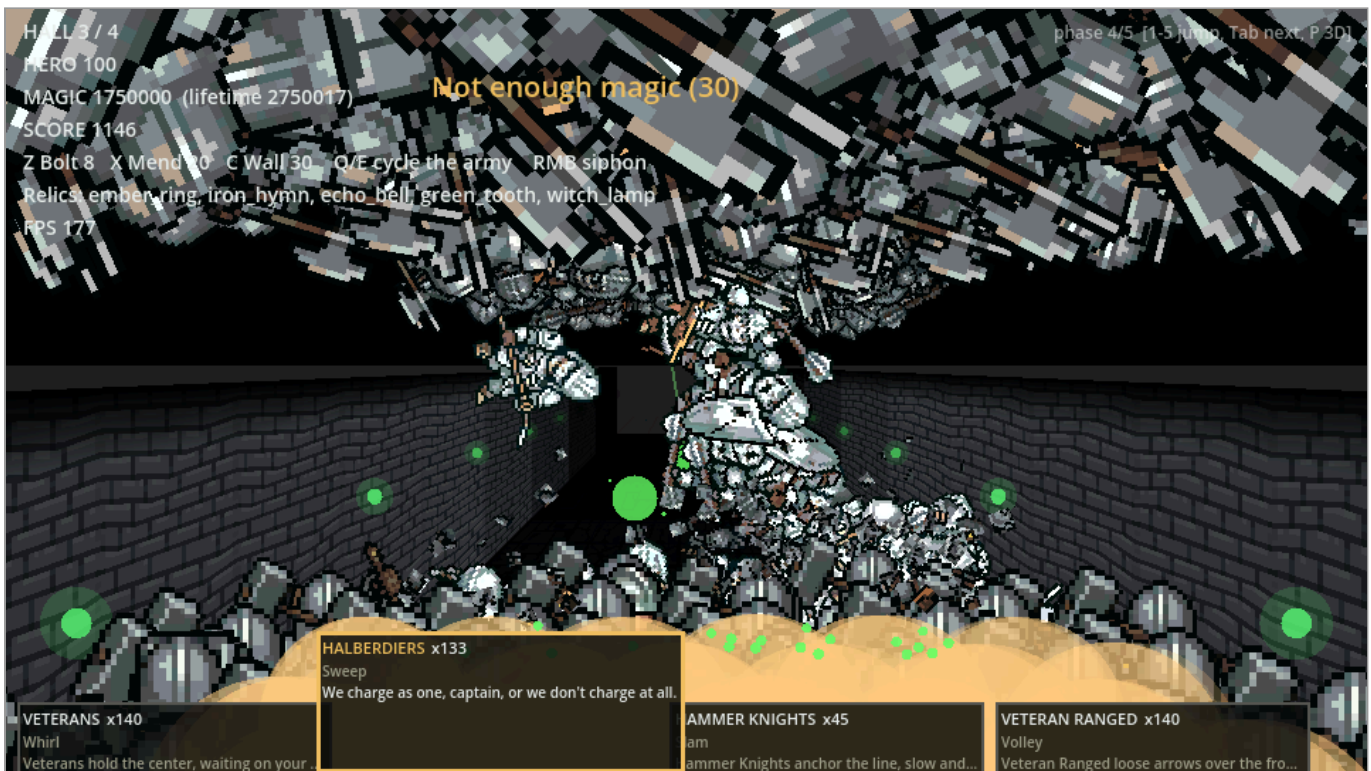
Bug 1 — outer file of the rank block drawn over the wall bricks, both sides (seed 7, t=1.5 s).



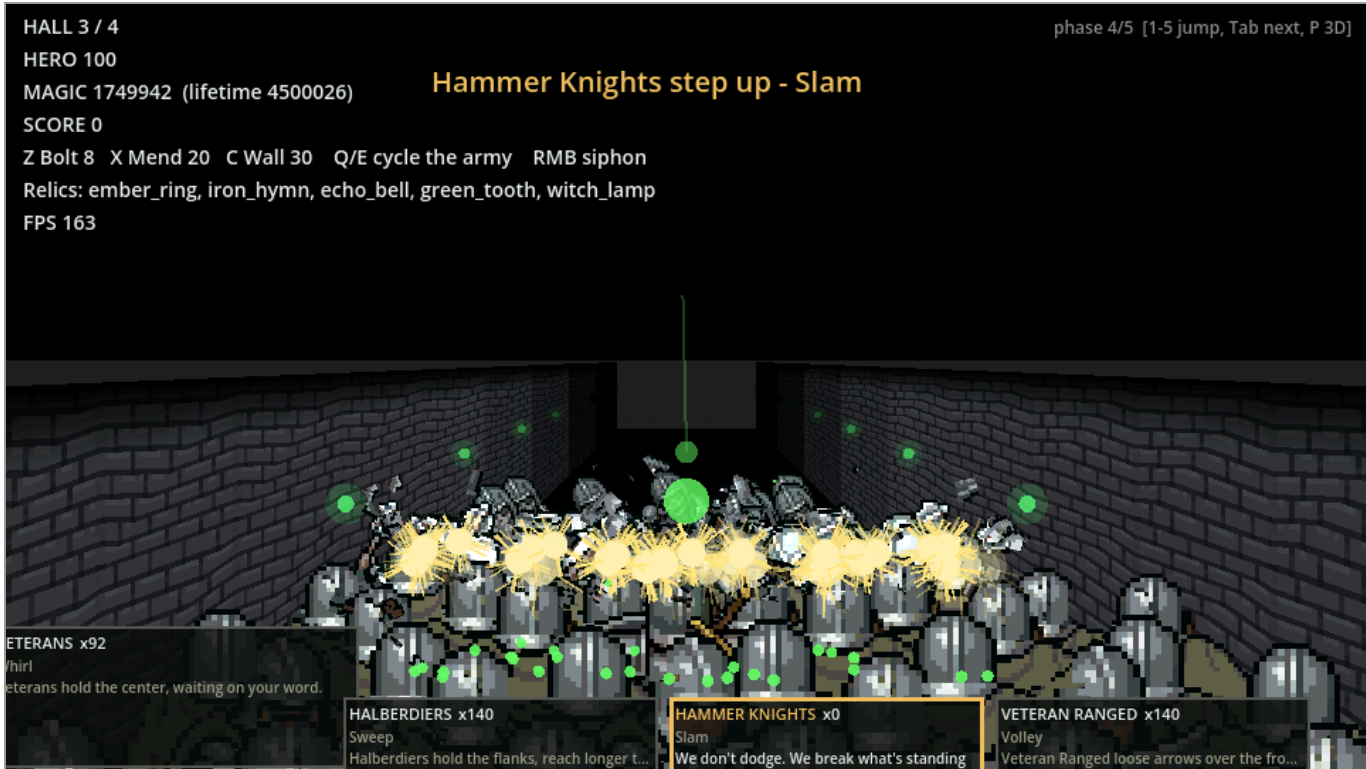
Bug 1 after a swap — halberdiers at |wx| 222 vs HALL_HALF 200 (seed 23).



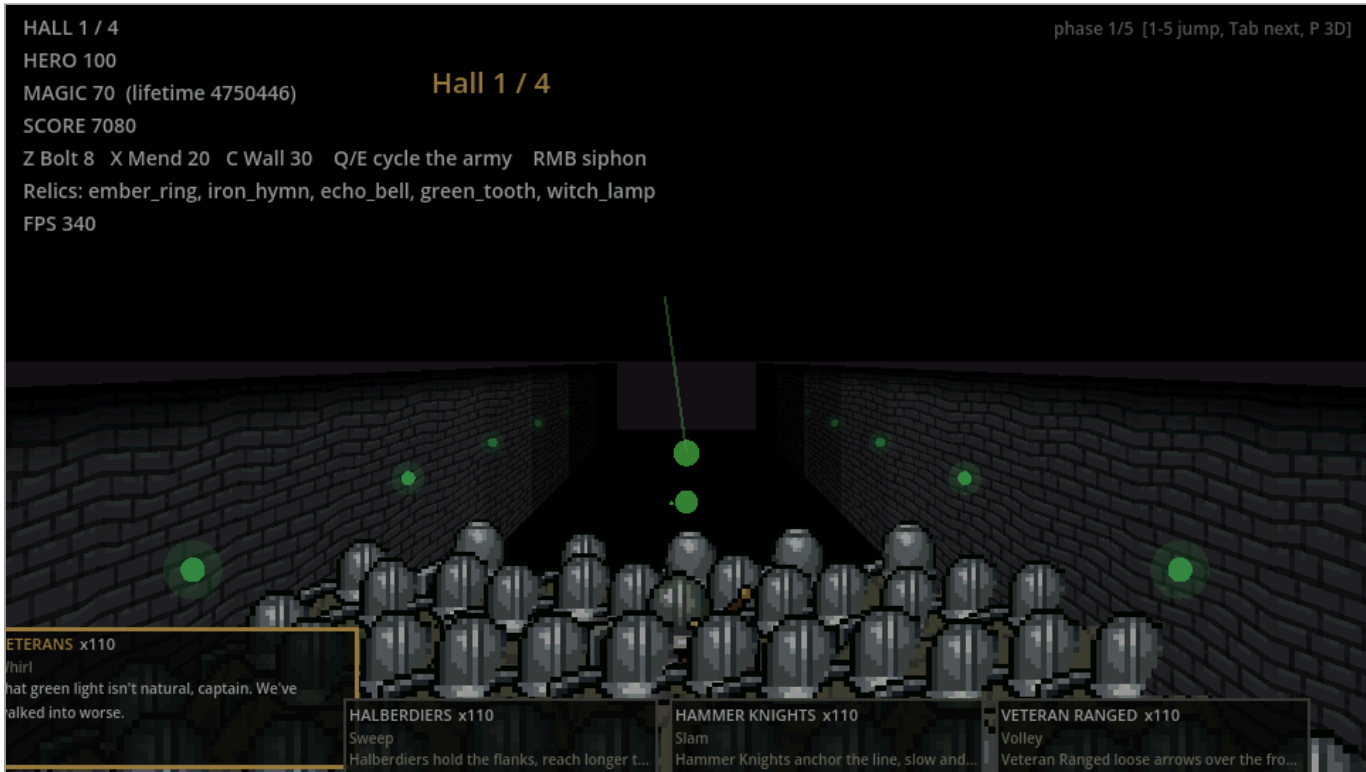
Bug 2 — the frame Engine.time_scale had been pinned at 0.05 for a full second (seed 11).



Bug 5 — both teams tumbling above the ceiling line 15 s after launch, hero untouched (seed 7, earlier run).



Final frame of seed 7: 4 hordes deep, wave 1 still ticking.



After the fixes — seed 23 final frame: the outer file now sits inside the bricks.

HALL 3 / 4

phase 4/5 [1-5 jump, Tab next, P 3D]

HERO 100

MAGIC 5249881 (lifetime 8000150)

SCORE 2167

Z Bolt 8 X Mend 20 C Wall 30 Q/E cycle the army RMB siphon

Relics: ember_ring, iron_hymn, echo_bell, green_tooth, witch_lamp

FPS 258



After the fixes — seed 7 final frame: a juggled crowd held inside the frame by Unit_air_cap().

Appendix — report excerpt (docs/qa/adversary_7.json)

```
{
  "run": {
    "behaviors": {
      "corner_hold": 3,
      "cursor_warp": 1,
      "horde": 5,
      "jump_phase": 4,
      "kill_front": 8,
      "kill_hero": 10,
      "launch_all": 5,
      "magic_drain": 4,
      "magic_flood": 4,
      "move_random": 5,
      "siphon_hold": 5,
      "spell_spam": 4,
      "swap_spam": 2,
      "teleport_hero": 9,
      "wall_hug": 6
    },
    "date": "2026-08-24T14:54:21",
    "invariants": [
      "hero clamp",
      "magic >= 0",
      "unit hp in 0..max",
      "dead flag matches hp",
      "front[] refs live + lane-consistent",
      "pool >= 0",
      "unit inside hall walls + depth range",
      "air_h >= 0",
      "no NaN",
      "transient states end within budget",
      "units[] < 1500",
      "fps >= 20",
      "Engine.time_scale back to 1 within 1s"
    ],
    "scene": "battle",
    "seconds": 60.0,
    "seed": 7
  },
  "findings": [
    {
      "count": 44001,
      "detail": "unit through the hall wall |wx| 217 > HALL_HALF 200",
      "error_type": "boundary_break",
      "game_context": {
        "army_type": "veteran",
        "behavior": "wall_hug",
        "enemies": 0,
        "fps": 70.0,
        "held_inputs": [
          "move_right"
        ],
        "hero_hp": 100.0,
        "magic": 0.0,
        "units": 48
      },
      "id": 1,
      "last_t": 60.0,
      "location": {
        "scene": "Battle",
```

```

    "state": "RANK",
    "system": "Battle.hall",
    "team": "ally",
    "unit_type": "veteran",
    "wave": 0,
    "wd": 259.0,
    "wx": 217.0
  },
  "screenshot": "adversary_7_f1.png",
  "t": 1.5
},
{
  "count": 268,
  "detail": "unit through the hall wall |wx| 218 > HALL_HALF 200",
  "error_type": "boundary_break",
  "game_context": {
    "army_type": "vet_ranged",
    "behavior": "swap_spam",
    "enemies": 30,
    "fps": 154.0,
    "held_inputs": [],
    "hero_hp": 100.0,
    "magic": 2.0,
    "units": 121
  },
  "id": 2,
  "last_t": 60.0,
  "location": {
    "scene": "Battle",
    "state": "ADVANCE",
    "system": "Battle.hall",
    "team": "ally",
    "unit_type": "vet_ranged",
    "wave": 0,
    "wd": 55.0,
    "wx": 218.0
  },
  "screenshot": "adversary_7_f2.png",
  "t": 17.6
},
"\u2026 2 more"
],
"engine_errors": [
  {
    "message": "ERROR: Lambda capture at index 2 was freed. Passed \"null\" instead.",
    "at": "at: call (modules/gdscript/gdscript_lambda_callable.cpp:242)",
    "count": 24
  },
  {
    "message": "ERROR: Lambda capture at index 3 was freed. Passed \"null\" instead.",
    "at": "at: call (modules/gdscript/gdscript_lambda_callable.cpp:242)",
    "count": 85
  },
  {
    "message": "SCRIPT ERROR: Invalid access to property or key 'team' on a base object of type 'Nil'.",
    "at": "at: (res://scripts/Battle.gd:332)",
    "count": 35
  },
  {
    "message": "ERROR: WASAPI: GetBufferSize error",
    "at": "at: thread_func (drivers/wasapi/audio_driver_wasapi.cpp:778)",
    "count": 1
  }
]
}

```

Appendix — CSV head (docs/qa/adversary_7.csv)

id	t	count	error_type	scene	wave	system	unit_type	team	state	wx	wd	behavior	magic	army_type	units	enemies	fps	detail	screenshot	
1	1.5	44001	boundary_break	Battle	0	Battle.hall	veteran	ally	RANK	217.0	259.0	wall_hug	0.0	veteran	48	0	70.0	unit through the hall wall wx 217 > HALL_HALF 200	adversary_7_f1.png	
2	17.6	268	boundary_break	Battle	0	Battle.hall	vet_ranged	ally	ADVANCE	218.0	55.0	swap_spam	2.0	vet_range	121	30	154.0	unit through the hall wall wx 218 > HALL_HALF 200	adversary_7_f2.png	
3	17.6	120	boundary_break	Battle	0	Battle.hall	halberdier	ally	ADVANCE	213.0	55.0	swap_spam	2.0	halberdie	r	168	29	154.0	unit through the hall wall wx 213 > HALL_HALF 200	adversary_7_f3.png